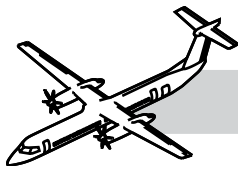


CHAPTER 76 ENGINE CONTROLS

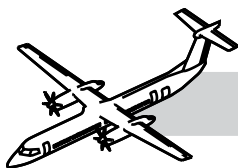
CONTENTS

	Page
76-00-00 INTRODUCTION	76-1
GENERAL	76-1
76-10-00 POWER CONTROL	76-3
General	76-3
System Description	76-3
Component Description.....	76-11
Controls and Indications.....	76-19
Operation	76-21
76-20-00 EMERGENCY SHUTDOWN.....	76-24
Introduction	76-24
General	76-24
System Description	76-24
76-00-00 MAINTENANCE PRACTICES.....	76-25



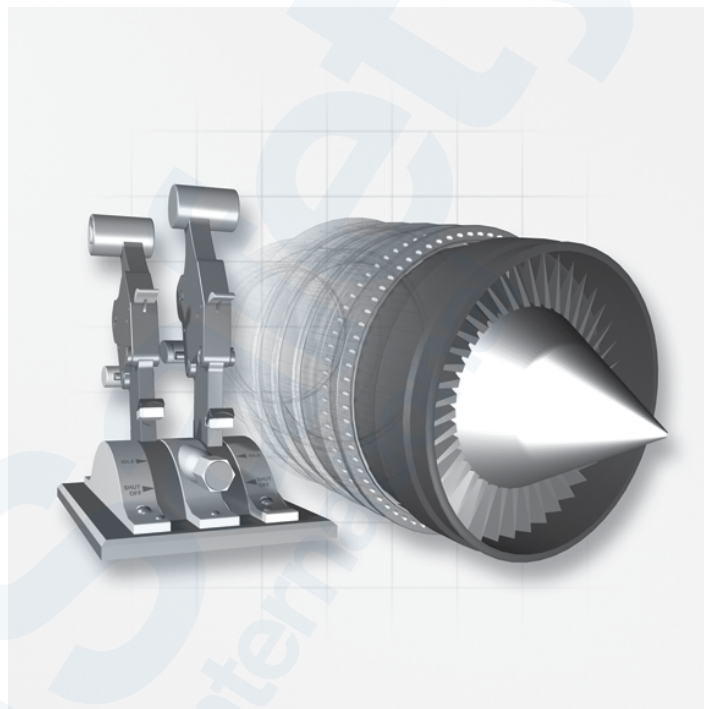
ILLUSTRATIONS

Figure	Title	Page
76-1	Pilot Control Pedestal and FPP	76-2
76-2	Power and Condition Lever Quadrant.....	76-4
76-3	NTOP Indications	76-6
76-4	MCL Indications.....	76-7
76-5	MCR Indications.....	76-8
76-6	START FEATHER Indications.....	76-9
76-7	Engine Cockpit Interface Unit (ECIU).....	76-10
76-8	ECIU Schematic	76-12
76-9	Engine Control Panel.....	76-14
76-10	MTOP Indications.....	76-16
76-11	RDC TOP Indications	76-18
76-12	Power Lever RVDT and Microswitches Schematic.....	76-20
76-13	Condition Lever RVDT and Microswitches Schematic.....	76-22



CHAPTER 76

ENGINE CONTROLS



76-00-00 INTRODUCTION

The engine controls provide:

- Power management and
- Propeller control.

The system is divided into two sub-systems:

- Power Control
- Emergency Shutdown.

GENERAL

The engines are controlled by the two power and the two condition levers.

Emergency shutdown stops the flow of fuel to the engine.

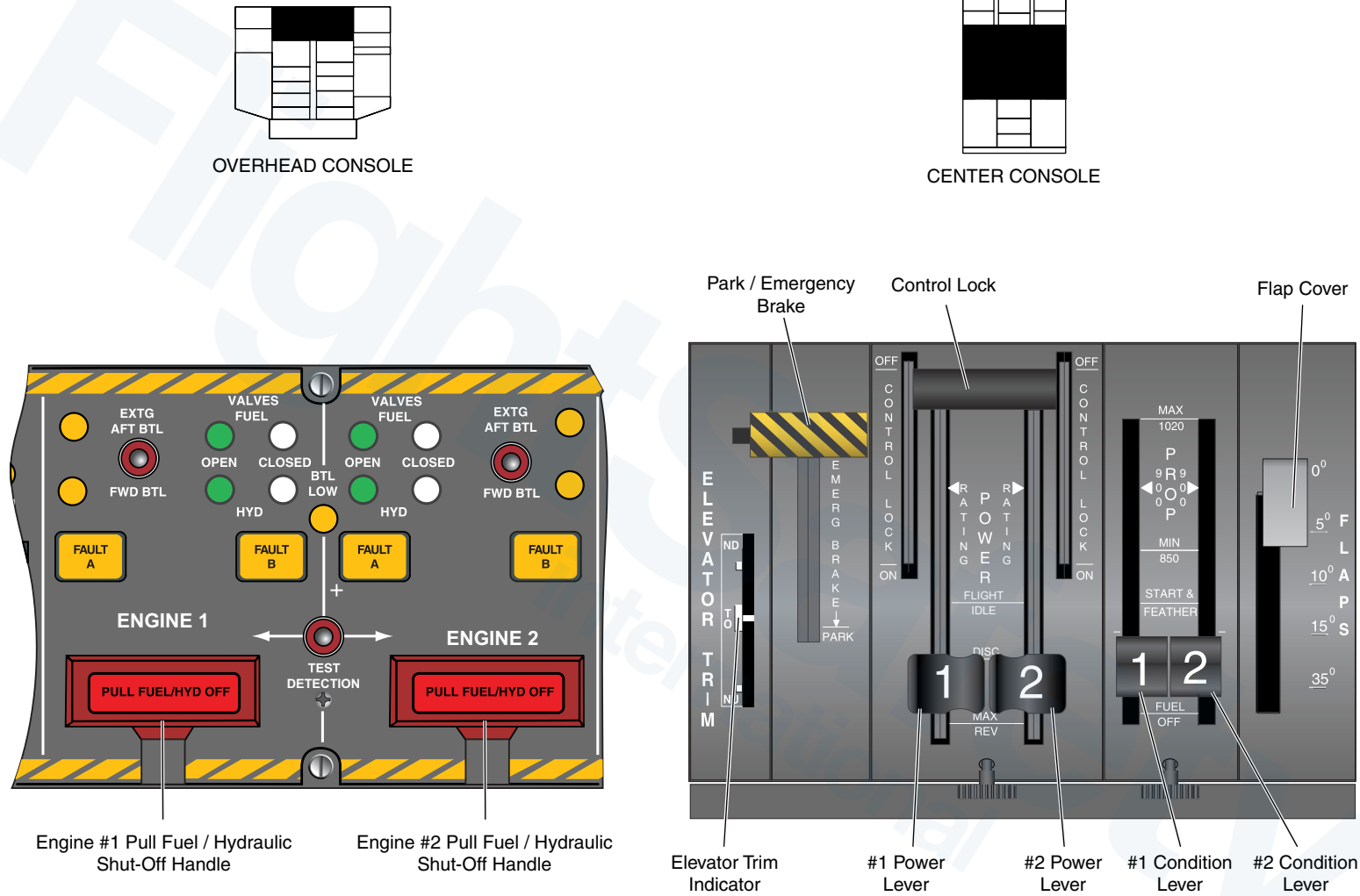
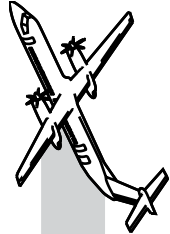
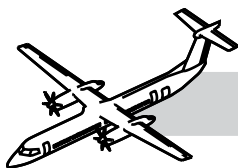


Figure 76-1. Pilot Control Pedestal and FPP





76-10-00 POWER CONTROL

GENERAL

Power management and propeller control is supplied to the aircraft through the power control system.

This system is comprised of the following components:

- Power Quadrant
- Condition Quadrant
- Engine Control Panel
- Pilot Control Pedestal
- ECIU
- Propeller Control Panel.

SYSTEM DESCRIPTION

Refer to Figure 76-1. Pilot Control Pedestal and FPP.

The FADEC and PEC electronically control the engine and propeller through:

- Power Lever Angle (PLA)
- Condition Lever Angle (CLA)
- Operating mode.

The FADEC and the PEC receive electrical signals from the Rotary Variable Differential Transducers (RVDT) and the micro-switches of the power and condition levers.

These electrical signals are computed by the FADEC and PEC.

The system also gives these power lever position discretes:

- Beta lockout warning
- Propeller Ground Beta Enable
- Roll spoiler Lift Dump
- Cabin Pressure Control System
- Auto-feather and Up-trim PLA high
- Oil Cooler Ejector Valve.

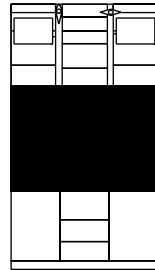
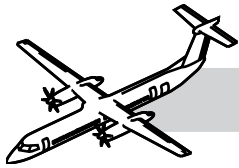
The system also gives these condition lever discretes:

- Engine Shutdown Switch
- Hydraulic Pump Caution Light
- Alternate Feather Lockout.

The PULL FUEL/HYD OFF handle is used to initiate an emergency shutdown of the engine.

When the handle is pulled, the fuel emergency shut off valve is energized to the closed position, shutting off the fuel supply. It will also energize a shutdown solenoid on the FMU.

Pulling the handle will also energize the emergency hydraulic shut-off valve to the closed position to shut off the hydraulic fluid supply to the EDP (Engine Driven Pump).



CENTER CONSOLE

76 ENGINE CONTROLS

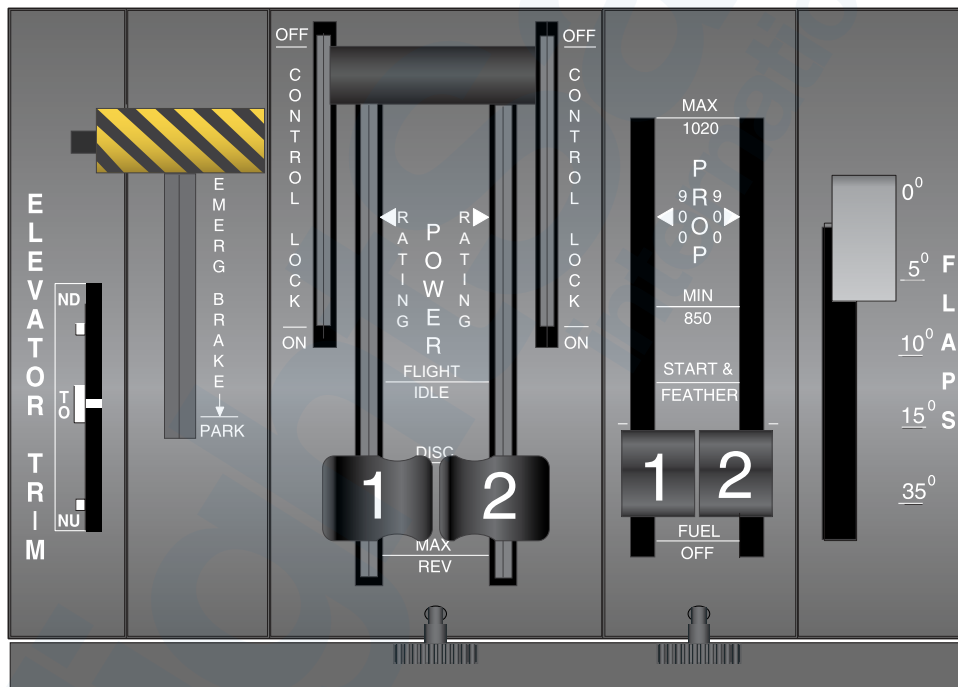
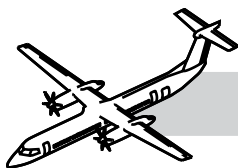


Figure 76-2. Power and Condition Lever Quadrant



Pilot Control Pedestal

Refer to Figure 76-2. Power and Condition Lever Quadrant.

The power and condition lever quadrant has the power levers, condition levers, flap lever, control lock lever, park/emergency brake lever and elevator trim indication.

A CONTROL LOCK handle is forward of the power levers.

When the handle is in the ON position the power levers cannot be advanced to the take off position.

A friction knob is below both the power and condition levers. Turning the knob in the FRICTION INCREASE direction progressively increases friction and resistance to movement. The friction load can be reduced by turning the knob in the opposite direction.

The Flap Lever is covered in detail in ATA 27-51-00.

The elevator trim indication is covered in detail in ATA 27-32-00.

The park/emergency brake is covered in detail in ATA 32-44-00.

Power Lever Quadrant

Power lever movement is limited by fixed stops at both ends of the quadrant. Each lever has these settings:

- RATING (rated power detent)
- FLIGHT IDLE (detent and gate)
- DISC (detent)
- MAX REVERSE (detent and stop).

RATING (80° ± 2.5° PLA)

With the power lever in the RATING (or rated power) detent, the engine will deliver the maximum horsepower demanded by the rating selections. The rating is selected by the Condition Lever Angle (CLA) or by the rating select switches on the engine control panel.

Each power lever has an over travel margin beyond the RATING detent (95 to 100 PLA), to allow the pilot select 25% extra power for emergency operation only.

FLIGHT IDLE (35° PLA)

Flight idle is the minimum position that the power lever can be set in flight. A Beta Lockout warning tone is generated by the warning tone generator when the:

- Radio altimeter output is more than 20 ft (6.1 m)
- Power lever trigger is lifted
- PLA is at flight idle or lower.

A flight idle gate for each power lever prevents inadvertent selection of engine ground beta and reverse operation in flight. The gate permits unrestricted forward movement without lifting the lever trigger. To move the lever aft of flight idle, the release trigger must be pulled.

DISC (20° PLA)

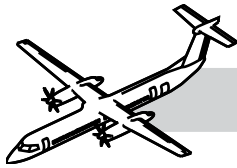
With the power lever in the DISC detent:

- The engine will supply minimum power
- The Propeller will have a -3.5 degree blade angle.

MAX REVERSE (0° to 5° PLA)

With the power lever moved rearward to the aft stop:

- The engine will supply a maximum of 1500 Shaft horsepower
- The propeller blade angle will move to the maximum reverse stop.



Condition Lever Quadrant

The condition levers are installed in the pilot's control pedestal, to the right of the power levers. The positions are marked on the quadrant to indicate:

- Propeller control range
- Engine start and propeller setting
- Fuel shut-off position.

Each condition lever has five distinct settings in the console slot. The condition lever detents are:

- MAX (1020)
- 900 RPM

- MIN (850)
- START/FEATHER
- FUEL OFF.

NTOP Indications

Refer to Figure 76-3. NTOP Indications.

MAX 1020

With the CLA in this position and the PLA in the RATING detent, Normal Take Off Power (NTOP) is demanded at a propeller speed of 1020 N_p.

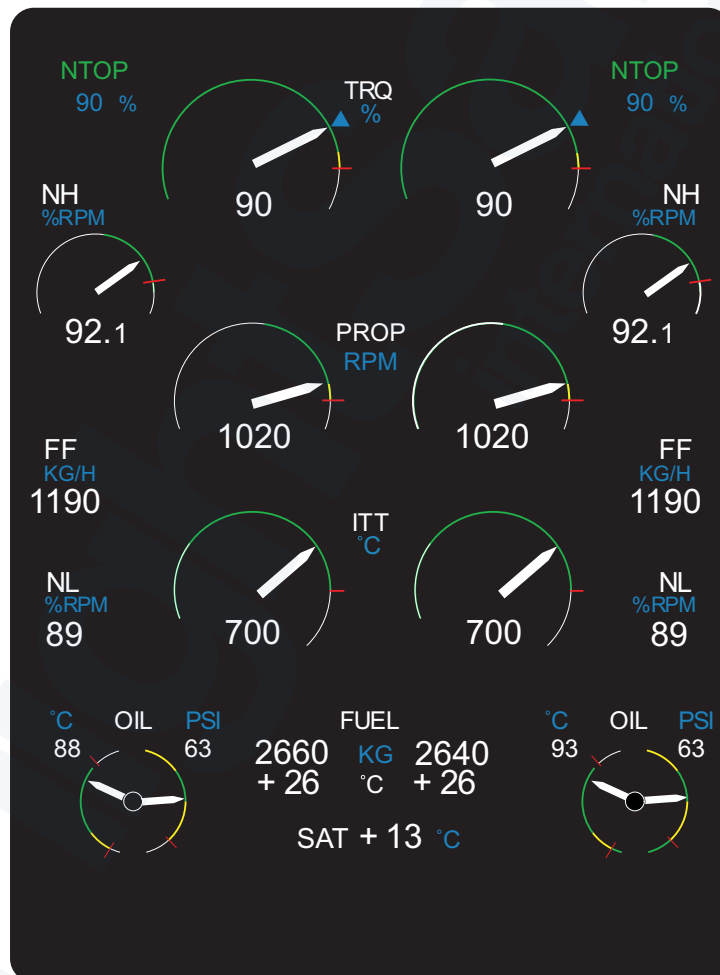
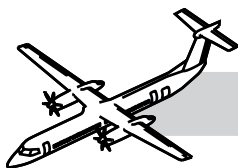


Figure 76-3. NTOP Indications



MCL Indications

NOTES

Refer to Figure 76-4. MCL Indications.

900 RPM

With the CLA in this position and the PLA in the RATING detent, Maximum Climb Power (MCL) is demanded at a propeller speed of 900 N_p.

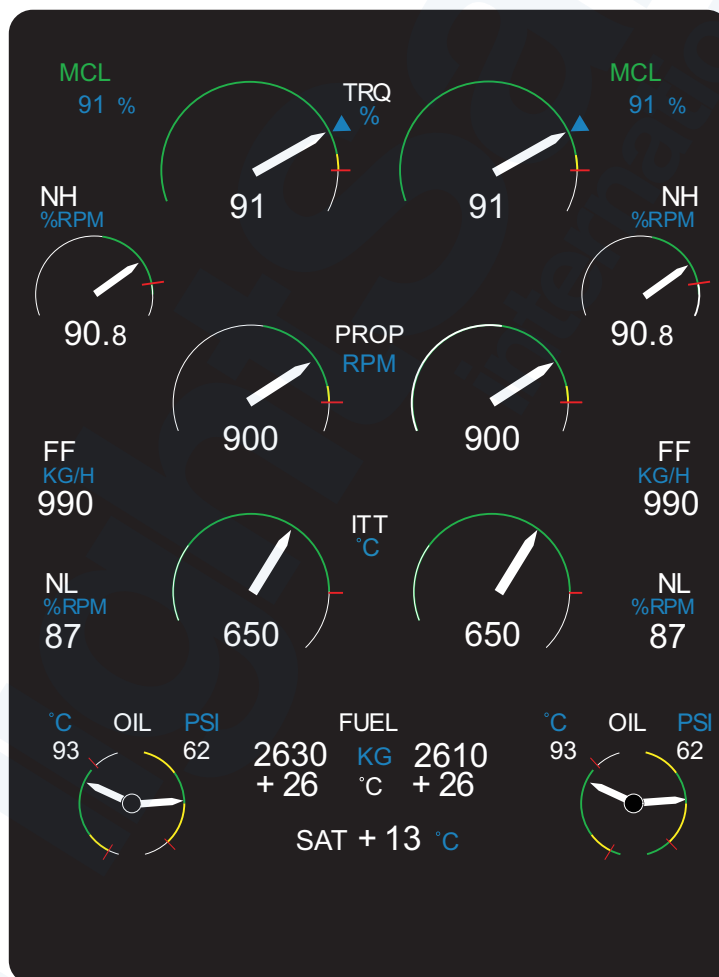
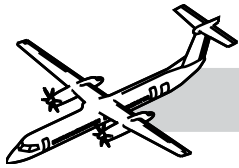


Figure 76-4. MCL Indications



MCR Indications

NOTES

Refer to Figure 76-5. MCR Indications.

MIN 850

With the CLA in this position and the PLA in the RATING detent, Maximum Cruise Power (MCR) at 850 N_P is demanded.

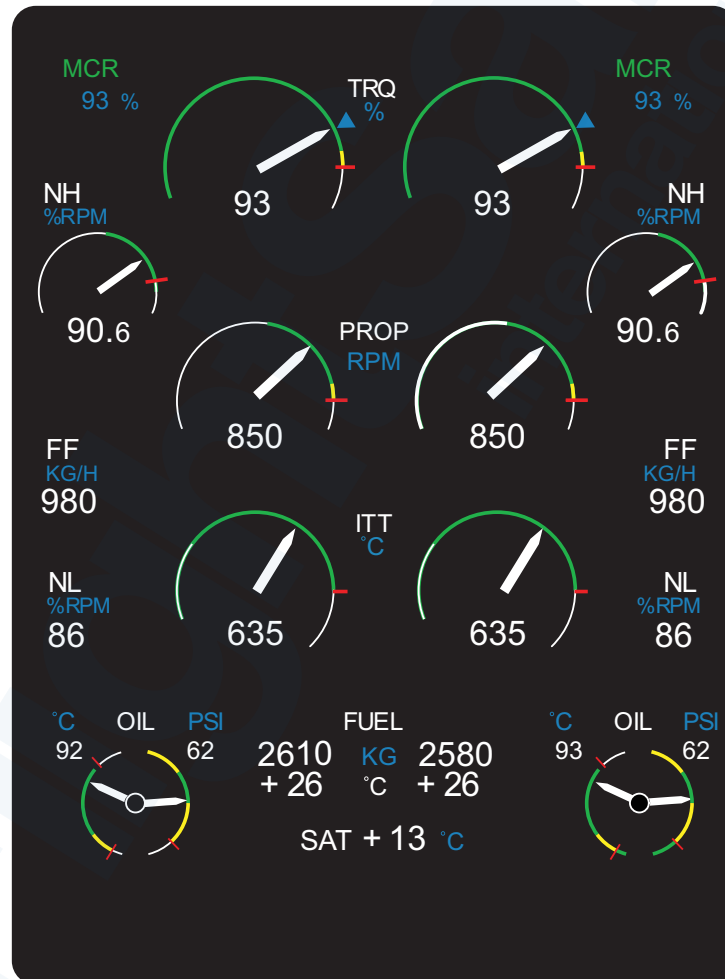
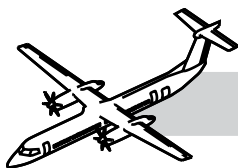


Figure 76-5. MCR Indications



START FEATHER Indications

Refer to Figure 76-6. START FEATHER Indications.

With the CLA in START FEATHER and the PLA in DISC, N_H is 64.2% and N_P is approximately 230 N_P . Torque will be approximately 11%.

FUEL OFF

With the CLA in this position, fuel to the engine is cut off.

There are two gates on the condition levers

that prevent inadvertent selection of propeller feather and fuel shut-off selections. The gates permit forward movement of the lever without lifting the lever. The condition lever must be lifted to allow it to pass the gate.

As the lever is moved rearward (toward the FUEL OFF position) it meets a gate at:

- The MIN position
- The START FEATHER position.

Lever movement at FUEL OFF is limited by a fixed stop at the rear of the quadrant.

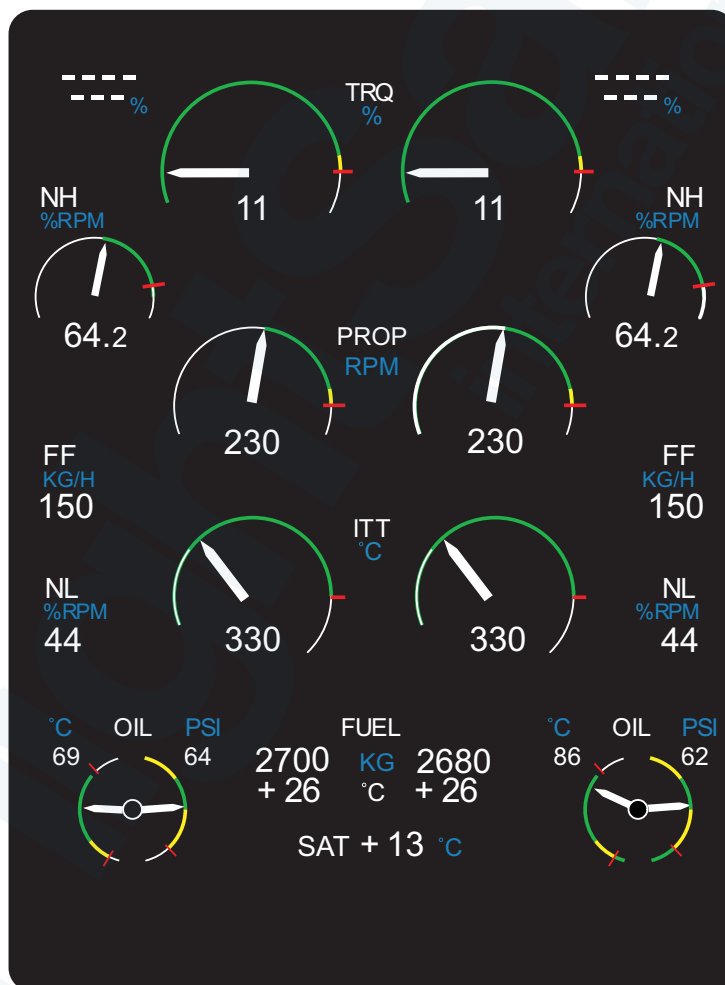


Figure 76-6. START FEATHER Indications

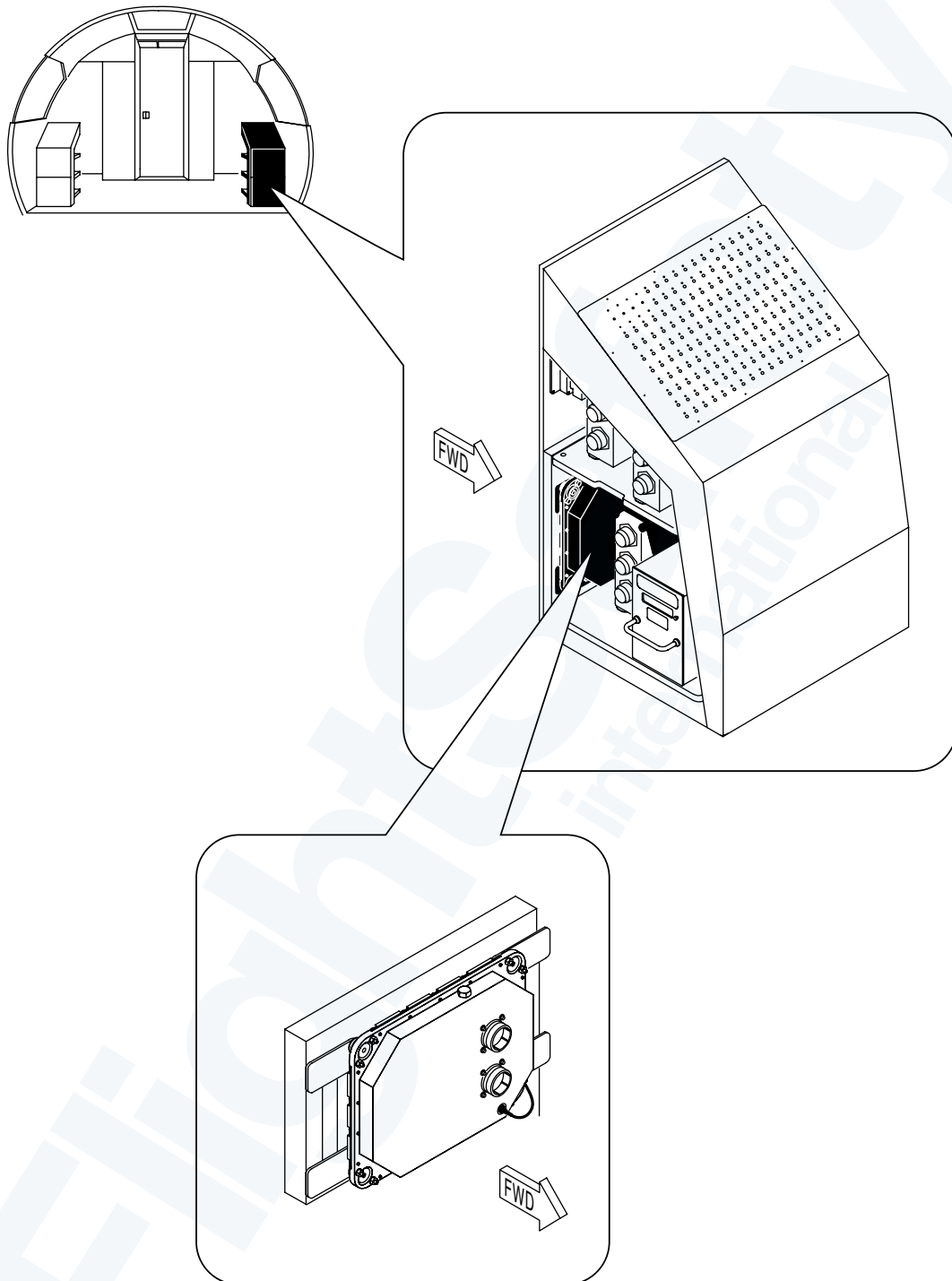
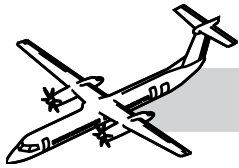
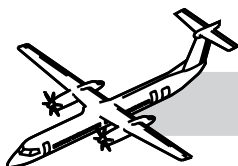


Figure 76-7. Engine Cockpit Interface Unit (ECIU)



COMPONENT DESCRIPTION

NOTES

Engine Cockpit Interface Unit (ECIU)

Refer to:

- Figure 76-7. Engine Cockpit Interface Unit (ECIU).
- Figure 76-8. ECIU Schematic.

The ECIU is installed vertically with anti-vibration mounts, below the pilot circuit breaker console (lower shelf).

The ECIU is a dual channel system that sends and receives ARINC 429 data from both FADEC's. The ECIU converts discrete analog inputs to digital outputs for insertion into the ARINC 429 data stream. Also, it converts discrete digital inputs to digital analog outputs.

The ECIU provides the following inputs and outputs:

- Engine Control Panel inputs to both FADEC's
- Environmental Control System (ECS) input to both FADEC's
- Ignition Control Panel input to both FADEC's
- Maintenance Panel input to both FADEC's
- FADEC outputs to the oil cooler flap door and ejector
- FADEC data exchange for T1.8 comparison and averaging
- FADEC outputs to the CAWP.

FADEC channel A of engine 1 communicates with FADEC channel B of engine 2 through ECIU channel A.

FADEC channel B of engine 1 communicates with FADEC channel A of engine 2 through ECIU channel B.

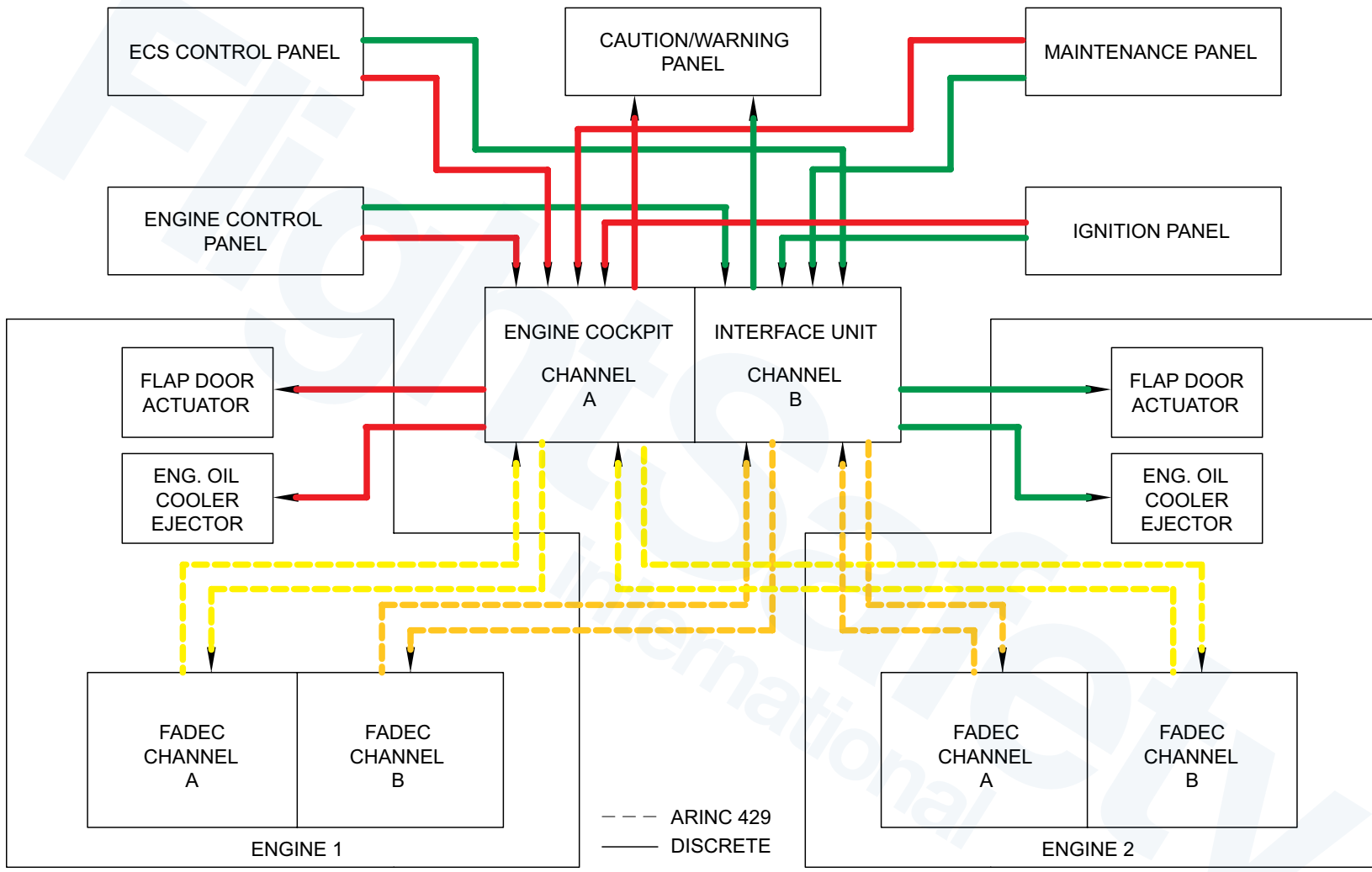
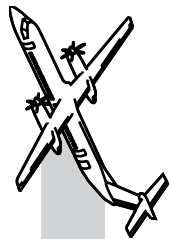
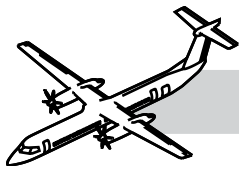


Figure 76-8. ECIU Schematic





PAGE INTENTIONALLY LEFT BLANK

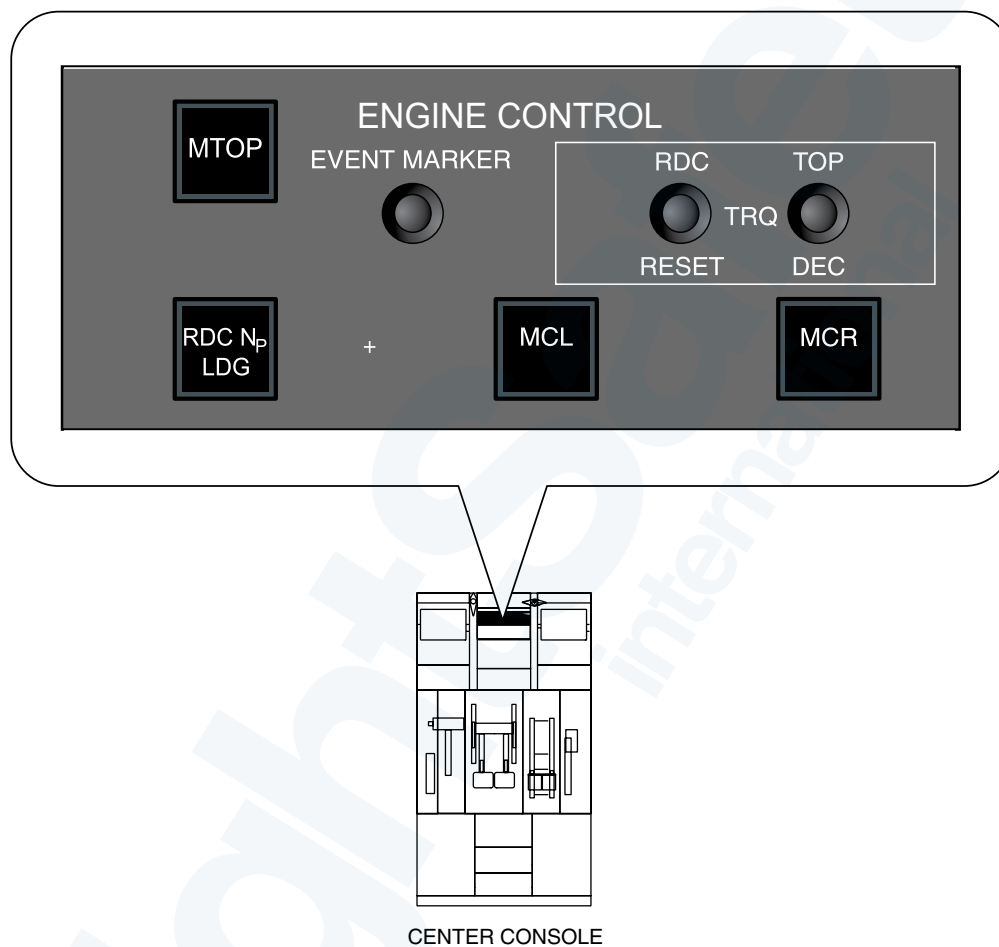
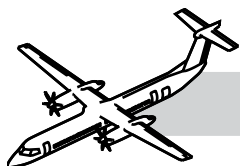
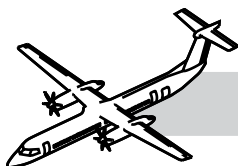


Figure 76-9. Engine Control Panel



Engine Control Panel

Purpose

The engine control panel provides alternate power rating selections.

Refer to Figure 76-9. Engine Control Panel.

The engine control panel is on the center console, forward of the power and condition levers. The panel has the following switches:

- Maximum Take-Off Power (MTOP) switch (optional)
- Reduced propeller RPM (RDC N_P) switch (optional)
- Maximum Climb Power (MCL) switch
- Maximum Cruise Power (MCR) switch
- Reduced Power Take Off (RDC TOP) DEC
- RESET switch
- EVENT MARKER switch.

Pushbutton and switchlight selections are transmitted as discretes to the ECIU. The ECIU converts the analog discretes to a digital signal and transmits them to the FADEC on an ARINC 429 bus.

The selected rating stays latched until another selection is made with either:

- CLA
- Pushbutton.

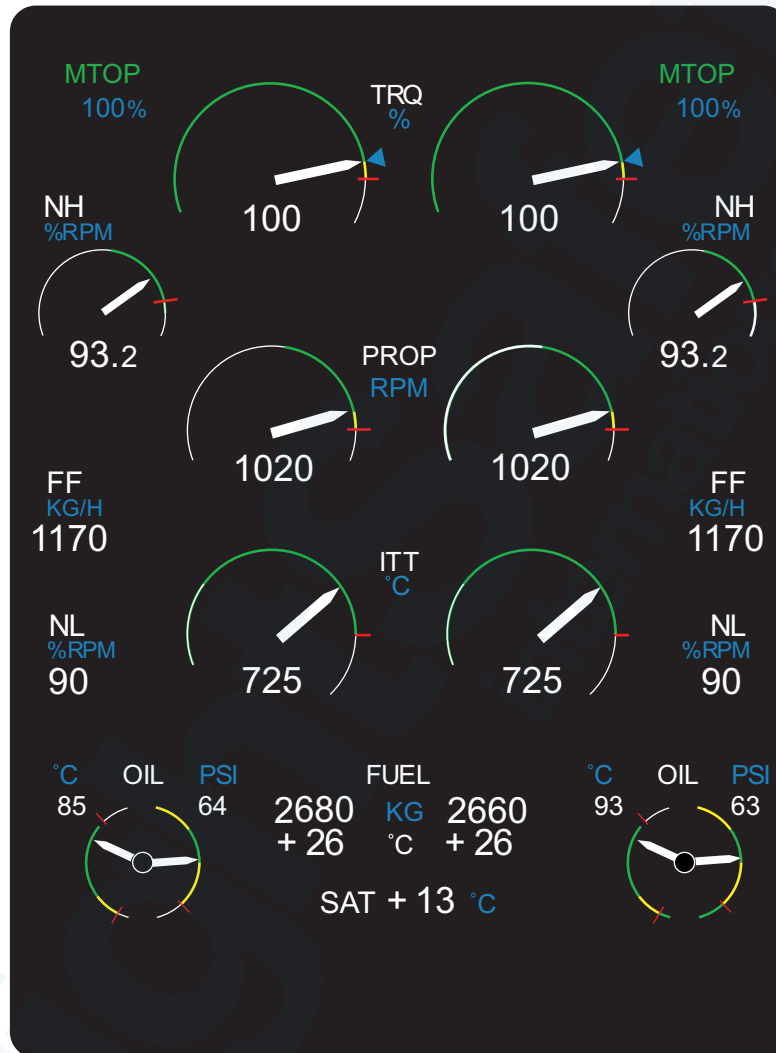
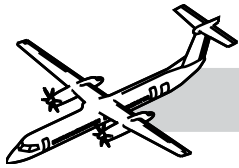
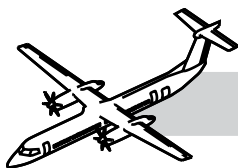


Figure 76-10. MTOP Indications



MTOP Switch (Optional)

Refer to Figure 76-10. MTOP Indications.

The MTOP switch selects maximum take off power request when:

- CLA is at MAX 1020
- MTOP discrete selected.

RDC N_p Switch (Optional)

The RDC N_p switch selects a reduced N_p for noise reduction during landing using logic in the FADEC. To activate RDC N_p for landing, all these conditions apply:

- CLA at 850 and
- PLA at less than 60° and
- Aircraft in the air.

Selecting the Reduced N_p switch will:

- Latch N_p at 850 and
- ED will show REDUCED N_p.

Pilot will move the CLA to MAX 1020, propeller speed will remain at 850, NTOP will be displayed. Propeller and power rating will advance to Max 1020 if:

- The switch is pushed again or
- PLA increases above 60°
- CLA NOT advanced to MAX 1020 position within 15 seconds of selecting RDC NP.

MCL Switch

When the condition lever is set to the 850 position and the MCL pushbutton switch is pushed, the MCR engine power rating usually set by the condition lever position changes to MCL. The MCR pushbutton switch is pushed to set the MCR engine power rating again. The MCL engine power rating also changes when the condition lever is moved.

MCR Switch

When the condition lever is set to 900 and the MCR pushbutton switch is pushed, the MCL engine power rating usually set by the condition lever position changes to MCR. The MCL pushbutton switch is pushed to set the MCL engine power rating again. The MCR engine power rating also changes when the condition lever is moved.

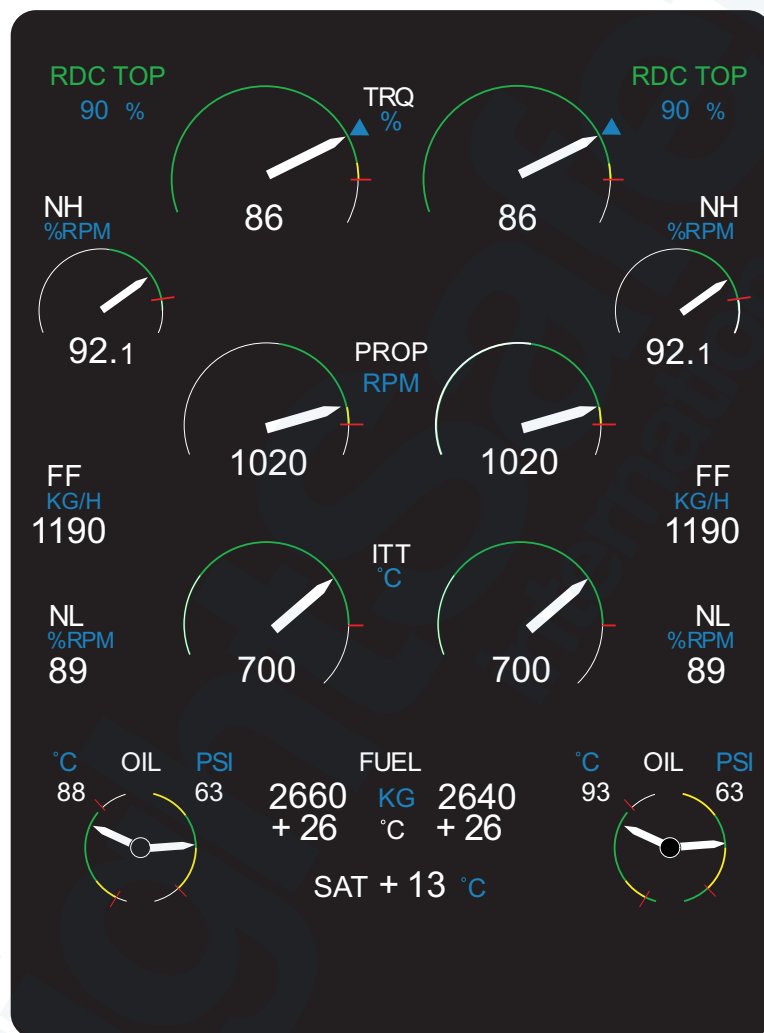
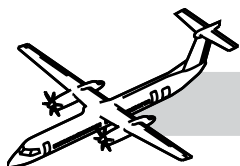
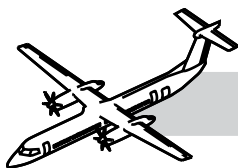


Figure 76-11. RDC TOP Indications



Reduced Take Off Pushbutton Switch

Refer to Figure 76-11. RDC TOP Indications.

The NTOP engine power rating can be decreased up to 10% in 2% increments for a reduced takeoff power (RDC TOP) rating when the conditions are as follows:

- The aircraft is on the ground
- The condition lever is set to MAX 1020 and the engine power rating is NTOP
- The power lever is set less than the RATING detent
- The RDC TOP DEC pushbutton switch is pushed (for a 2% decrease)
- The NTOP engine mode message changes to RDC TOP.

EVENT MARKER Switch

Instructs the EMU to record a snapshot and a 3 minute trace (2 minutes before, 1 minute after the selection).

Propeller Control Panel

The propeller control panel provides autofeather and alternate feather selection.

The propeller control panel is on the center console (forward of the power and condition levers). The panel has the two guarded alternate feather pushbuttons and one autofeather pushbutton.

More detail of the panel will be found in ATA 61-20-41.

CONTROLS AND INDICATIONS

The warning light outputs from the ECIU are as follows:

- No.1 ENG OIL PRESS
- No.2 ENG OIL PRESS.

The caution light outputs from the ECIU are:

- No.1 ENG FUEL PRESS
- No.2 ENG FUEL PRESS
- No.1 FUEL FLTR BYPASS
- No.2 FUEL FLTR BYPASS.

The discrete outputs from the ECIU are:

- No.1 engine Flap Door Actuator A
- No.1 engine Flap Door Actuator B
- No.2 engine Flap Door Actuator A
- No.2 engine Flap Door Actuator B
- No.1 engine Oil Cooler Ejector
- No.2 engine Oil Cooler Ejector.

Flight compartment discrete signals are processed by both ECIU channels to improve the availability of the signals.

RVDT	SWITCH	SWITCH	SWITCH	RVDT	SWITCH	SWITCH	SWITCH
0° TO 100° RIGHT PLA	RIGHT GO AROUND	RIGHT TRIGGER	RIGHT 33° PLA	0° TO 100° RIGHT PLA	RIGHT 47° PLA	RIGHT 60° PLA	RIGHT 60° PLA
FADEC Right Channel A	Integrated Flight Cabinet	Beta Lockout Warning Grounded	28 VDC GBE Time Delay Relay	FADEC Right Channel B	FCS ECU	PEC Right Channel	PSEU Aft Safety Valve OFC Control

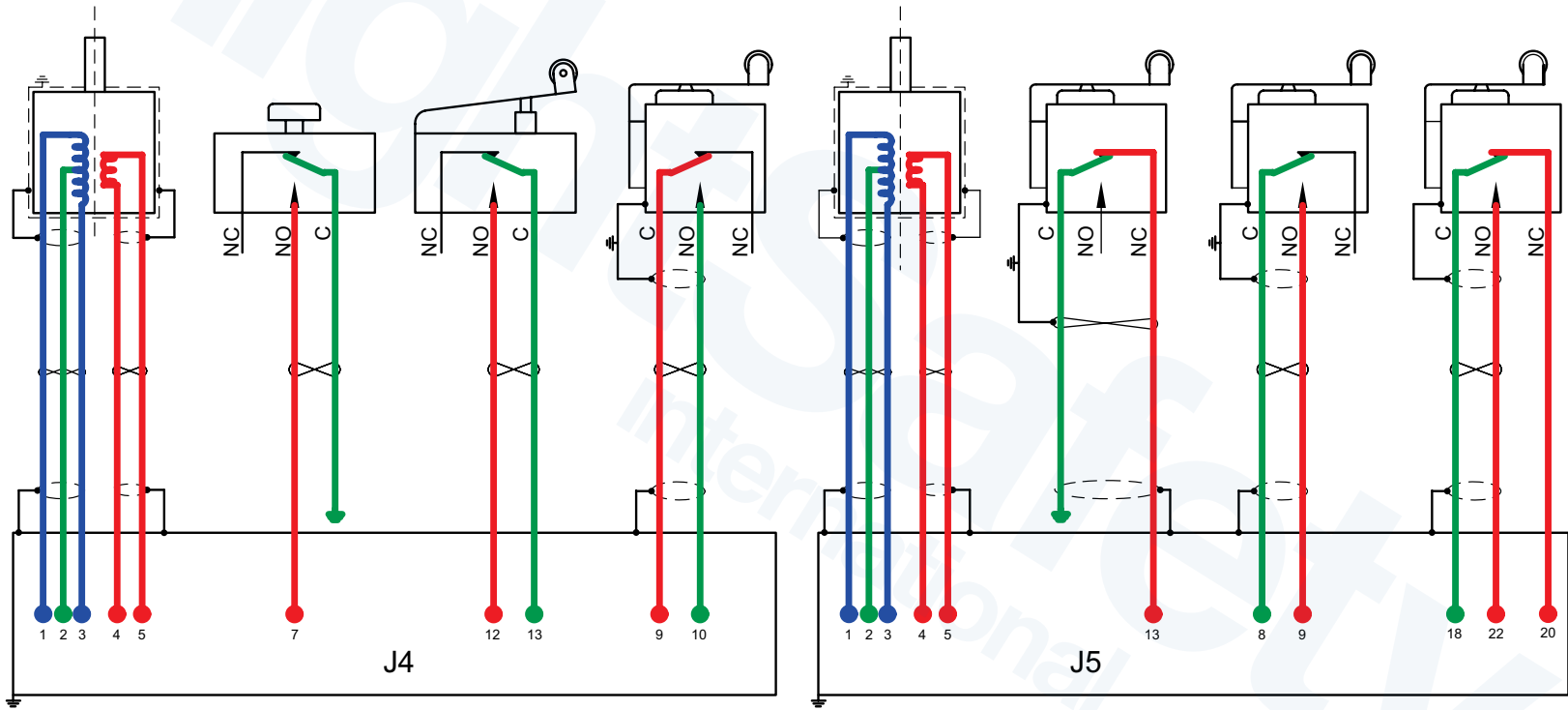
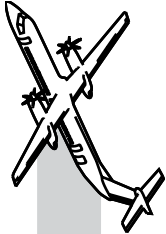
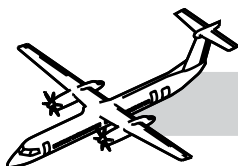


Figure 76-12. Power Lever RVDT and Microswitches Schematic





OPERATION

NOTES

Power Lever Quadrant

Refer to Figure 76-12. Power Lever RVDT and Microswitches Schematic.

Each power lever drives two identical RVDT, one for each FADEC channel.

The FADEC provides the excitation for the RVDT. Power lever position is represented by a proportional AC voltage at the FADEC.

Four microswitches are operated by the power lever, one at each of the following positions:

- PLA at 33° Ground Beta Enable Switch for propeller control in Discing and Reverse
- PLA at 47° Spoiler Lift Dump Switch armed
- PLA at 60° or above high PLA input for Autofeather Arming system
- PLA at 60° or above for pre-pressurization function in Cabin Pressure Control System (No.2 PLA only).

TYPE	RVDT	SWITCH	SWITCH	SWITCH	RVDT	SWITCH
DESIGNATION	0° TO 95° CLA	LEFT 15° CLA ENGINE START/SHUTDOWN	LEFT 15° CLA ENGINE START/SHUTDOWN	LEFT 15° CLA HYD EDP OIL FLAP DOOR	0° TO 95° LEFT CLA	LEFT 40° CLA ALTERNATE FEATHER LOCKOUT
TYPE		CLA AT 15°	CLA AT 15°	CLA AT 15°		CLA AT 40°
INTERFACE	PEC LEFT CHANNEL A	FADEC LEFT CHANNEL A	FADEC LEFT CHANNEL B	#1 ENG HYD PUMP CAUTION OIL FLAP DOOR TEST	PEC LEFT CHANNEL B	PEC + MANUAL UNFEATHER SWITCH + TIME DELAY RELAY ALTERNATE FEATHER SWITCH

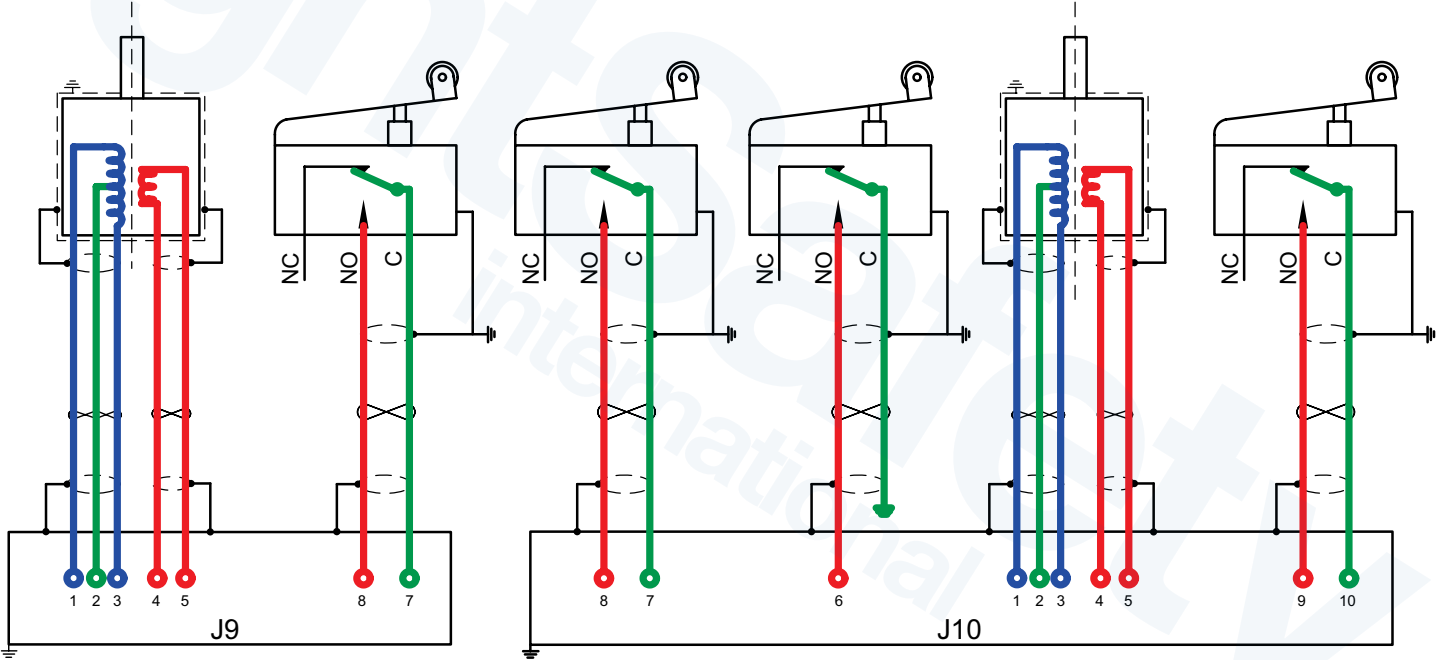
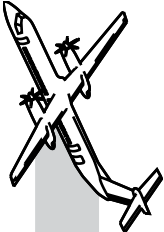
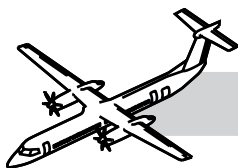


Figure 76-13. Condition Lever RVDT and Microswitches Schematic





Condition Lever Quadrant

NOTES

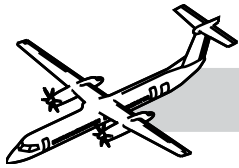
Refer to Figure 76-13. Condition Lever RVDT and Microswitches Schematic

Each condition lever drives two identical RVDT. Both RVDT's send Condition Lever Angle (CLA) signals to the PEC.

Four switches are operated by each condition lever at the following positions:

- CLA at 15° or below - Engine Start/Shutdown Switch send fuel off signals to the FADEC Channel A
- CLA at 15° or below - Engine Start/Shutdown Switch send fuel off signals to the FADEC Channel B
- CLA at 15° or below - Engine Hydraulic Pump Caution Switch, ensures light on when propeller feathered
- CLA at 40° or above - high CLA inhibits alternate feather function through Alternate Feather Lockout Switch.

The PEC provides the excitation for the RVDT. Condition lever position is represented by a proportional AC voltage back to the PEC.



76-20-00 EMERGENCY SHUTDOWN

NOTES

INTRODUCTION

The emergency shutdown system provides a quick (and safe) method to shut off engine fuel and hydraulic fluid in an emergency.

GENERAL

The following components are supplied for each engine:

- A PULL FUEL OFF handle
- A fuel emergency shut-off valve
- Fuel shut-off Solenoid (FMU)
- A hydraulic emergency shut-off valve
- Relays
- Circuit protection
- Position indication.

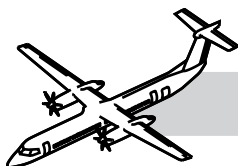
SYSTEM DESCRIPTION

Operation of the PULL FUEL/HYD OFF handle closes the related engine:

- Fuel shut-off valve
Refer to ATA 28-21-26 for details
- Hydraulic shut-off valve
Refer to ATA 28-11-06 for details
- Fuel shut-off Solenoid (FMU)
Refer to ATA 73-20-00 for details.

Refer to 26-11-00 in ATA 26 Fire Protection for more details.

It also closes a dedicated fuel shut-off solenoid valve at the Fuel Metering Unit. Refer to 73-20-00 in ATA 73 Fuel System for more details.



76-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 76-10-01-000-801: Removal of the Power Quadrant.
- AMM 76-10-01-400-801: Installation of the Power Quadrant.
- AMM 76-10-06-000-801: Removal of the Condition Quadrant.
- AMM 76-10-06-400-801: Installation of the Condition Quadrant.
- AMM 76-10-11-710-801: Operational Test of the Engine Control Panel.